UNIVERSAL BOUNDEDNESS: EXISTENCE OF ONE BOUNDED ETALE COVER VERSUS BOUNDEDNESS ON EVERY AFFINE ETALE MORPHISM

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Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Etale morphisms are open. Remark remark-recall (same chapter, structural). The etale maps q_i are open, so their images cover U.
- Bounded fibres over quasi-compact source. Morphisms, lemma locally-quasi-finite-qc-source-universally-bounded (cross-chapter, named). Fibres of p_{i_j} over the quasi-compact opens are universally bounded.
- Composition of universal boundedness. lemma-composition-universally-bounded (same chapter, carded L26049). Lifts boundedness from W_j -> U_{i_j} to W_j -> X.
- Permanence under etale surjection. lemma-universally-bounded-permanence (same chapter, carded L26046). Transfers the bound from the coproduct cover to U -> X.

Source reference. Original lemma 03JM: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 297–297. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03JP: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 384–384. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03JA: focused-seeds/geometry/S9-decent-u-universally-bounded/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 03II: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 459–459. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. The following are equivalent:*

(1) *there exist schemes U_i and étale morphisms $U_i \to X$ such that $\coprod U_i \to X$ is surjective and each $U_i \to X$ has universally bounded fibres, and*
(2) *for every affine scheme U and étale morphism $\varphi : U \to X$ the fibres of $U \to X$ are universally bounded.*

Proof. The implication (2) $\Rightarrow$ (1) is trivial. Assume (1). Let $(\varphi_i : U_i \to X)_{i \in I}$ be a collection of étale morphisms from schemes towards X, covering X, such that each φ_i has universally bounded fibres. Let $\psi : U \to X$ be an étale morphism from an affine scheme towards X. For each i consider the fibre product diagram

$$\begin{array}{ccc} U \times_X U_i & \xrightarrow{p_i} & U_i \\ {\scriptstyle q_i}\downarrow & & \downarrow{\scriptstyle \varphi_i} \\ U & \xrightarrow{\psi} & X \end{array}$$

Since q_i is étale it is open (see Remark 03II). Moreover, we have $U = \bigcup \mathrm{Im}(q_i)$, since the family $(\varphi_i)_{i \in I}$ is surjective. Since U is affine, hence quasi-compact we can finite finitely many $i_1, \ldots, i_n \in I$ and quasi-compact opens $W_j \subset U \times_X U_{i_j}$ such that $U = \bigcup p_{i_j}(W_j)$. The morphism p_{i_j} is étale, hence locally quasi-finite (see remark on étale morphisms above). Thus we may apply Morphisms, Lemma 03JA to see the fibres of $p_{i_j}|_{W_j} : W_j \to U_{i_j}$ are universally bounded. Hence by Lemma 03JM we see that the fibres of $W_j \to X$ are universally bounded. Thus also $\coprod_{j=1,\ldots,n} W_j \to X$ has universally bounded fibres. Since $\coprod_{j=1,\ldots,n} W_j \to X$ factors through the surjective étale map $\coprod q_{i_j}|_{W_j} : \coprod_{j=1,\ldots,n} W_j \to U$ we see that the fibres of $U \to X$ are universally bounded by Lemma 03JP. In other words (2) holds. □